

Project 1 - Poster Deliverable 48W x 36H inches (With Supporting R Scripts)

Poster delivered as PDF file, DO NOT PRINT

Objectives - Prepare your data for time series analysis and perform cursory data observations.

Week 1 - Learning Objectives	Deliverables	Grading
*Students can design a poster that effectively communicates key ideas using graphs, figures, and minimal text. *Students are able to identify the essential parts of an academic poster.	*A Visually Appealing 48W x 36H inch Poster pdf file (DO NOT PRINT POSTER) *Problem Statement and Importance *Solution Approach	*Poster is visually appealing /5 *Poster is well organized /5 *Poster text is kept minimal, short and easy to read, quickly communicating main points /5
The poster should contain:		
*Students can plot time series data using statistical software at different frequencies and timescales *Students can identify seasonalities in the data through observing time series plots	*time series plots for A1A2A3 Demand data that demonstrate different seasonalities present *time series plots for solar data that demonstrate different seasonalities present *time series plots for net demand data that demonstrate different seasonalities present *observations	*A1A2A3 plot(s) present /5 *A1A2A3 plot has readable x and y axis labels /2 *solar plot(s) present /5 *solar plot has readable x and y axis labels /2 *net demand plot(s) /5 *net demand plot has readable x and y axis labels /2 *observations for each plot included and accurate /10
*Students can prepare a data set for time series analysis. Including standardizing the time increments, imputing missing data, and removing anomalous data values. *Students can communicate when various imputation methods should be used. *Students can communicate methods to identify anomalous values.	*discussion of the extent of cleansing necessary (how many anomalous observations, missing observations, etc...) *discussion of imputation methods chosen, and graphs showing examples of missing/anomalous data with corresponding imputed values	*cleansing details provided /5 *imputation method provided /5 *imputation examples provided /10
*Students can apply data smoothers to time series data. *Students can communicate the effect of data smoothers on the time series data.	*Time series plot demonstrating the effect of data smoothers *Discussion of observations	*demonstration of smoothing in plot /5 *observation accurate /3
Week 2 - Learning Objectives		
*Students should be able to use differencing to remove the trend and seasonality in a data set.	*Plot of data with trend removed (net demand only) *Plot of data with seasonality removed (net demand only) *plot of data with both seasonality and trend removed (net demand only) *Comment on the appearance of the transformed dataset (does it appear to have constant variance and constant mean?)	*plot removing trend /3 *plot removing seasonality /3 *plot removing both trend and seasonality /3 *comment on data /5 *plots have labelled x and y axis /3
*Students should be able to calculate and interpret the mean and variance of a dataset *Students should be able to produce an ACF plot of data and identify if any autocorrelation is present in the data.	*Sample Variance (demand data, solar data, and net demand data) *Sample Mean (demand data, solar data, and net demand data) *Sample ACF Plot (net demand data), consider removing seasonality BEFORE creating ACF plot *Discussion of observations	*variances provided correctly /3 *means provided correctly /3 *ACF plots provided /6 *observations of ACF plots and statistics /4
Additional Project 1 Deliverables		
*1 or more R scripts that ingests the original CSV files, cleans the data, and (optionally) produces new cleansed data files *The R script should be configurable such that you can easily produce 5 minute interval data, 30 minute interval data, or 1 hour interval data		*reproducible R scripts that run and clean data /15 *scripts can be configured to aggregate by time interval /10

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